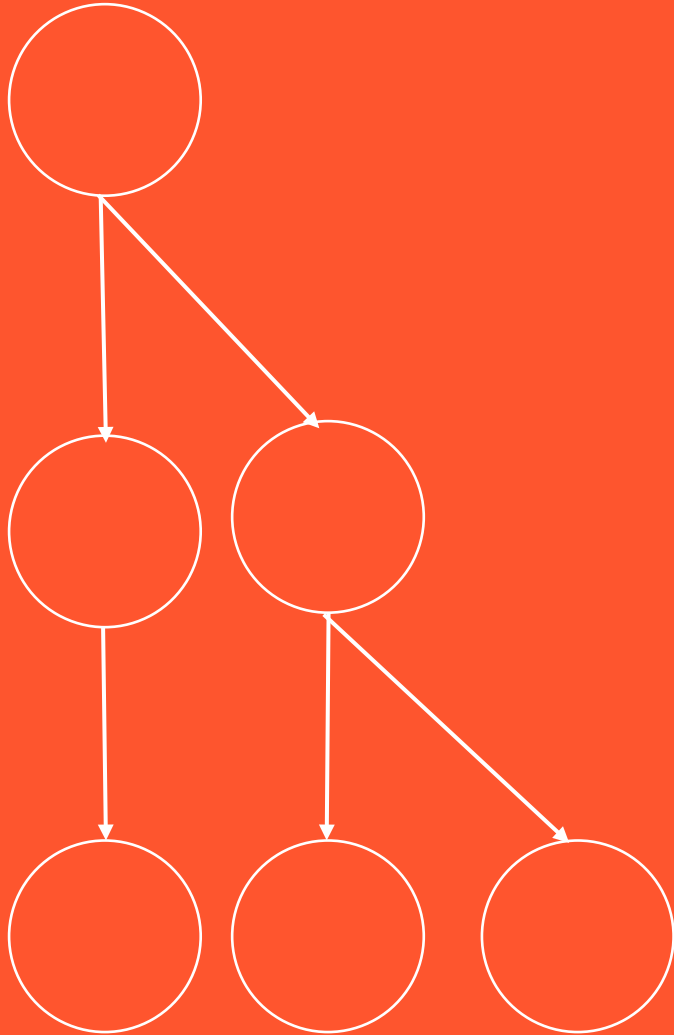


BCOG 100

Social Cognition

Lecture 1: The Evolution of Social Cognition in Primates



UNIVERSITY OF
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URBANA-CHAMPAIGN

Reading Check 10-1

Password: ToM

Question: What is theory of mind, and what's at least one way scientists have studied theory of mind

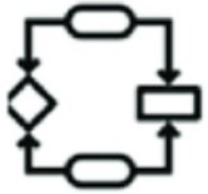


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Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

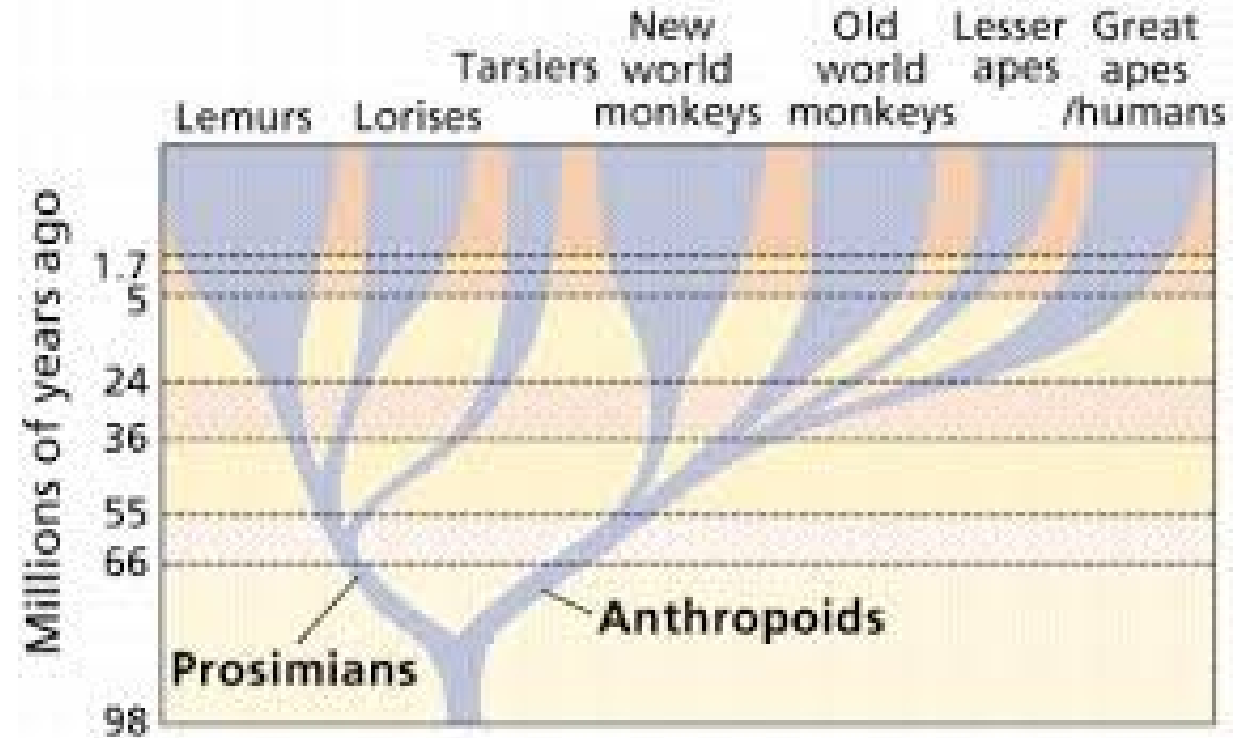
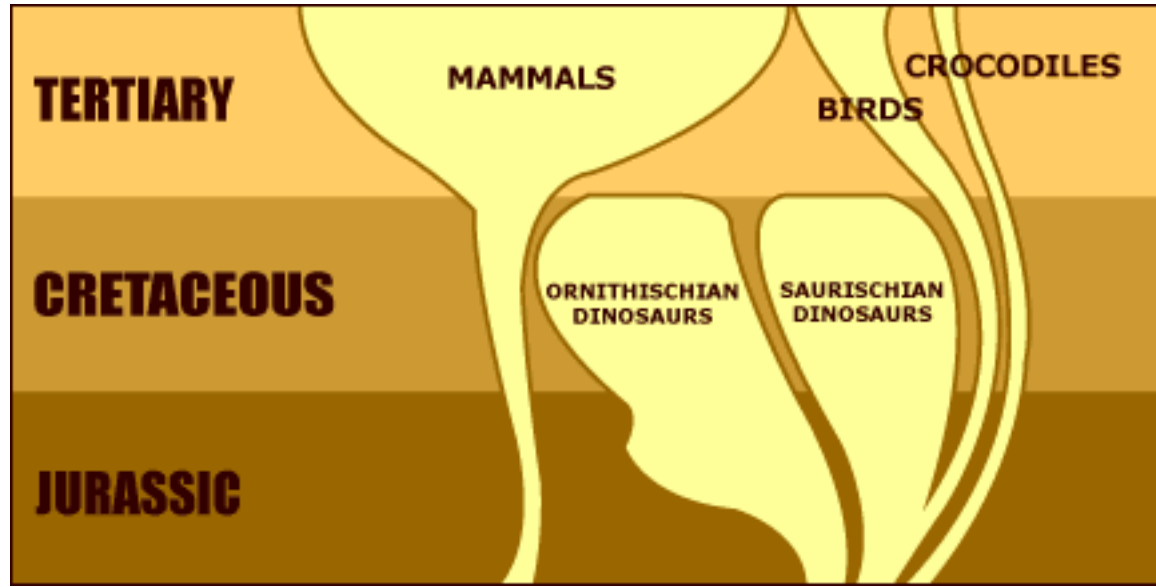
Last Time:

The evolution of the neocortex, memory, and generative models in early mammals.

Today:

- Some mammals start living in big, complex social groups
- A lot of new behaviors and abilities emerged as a result

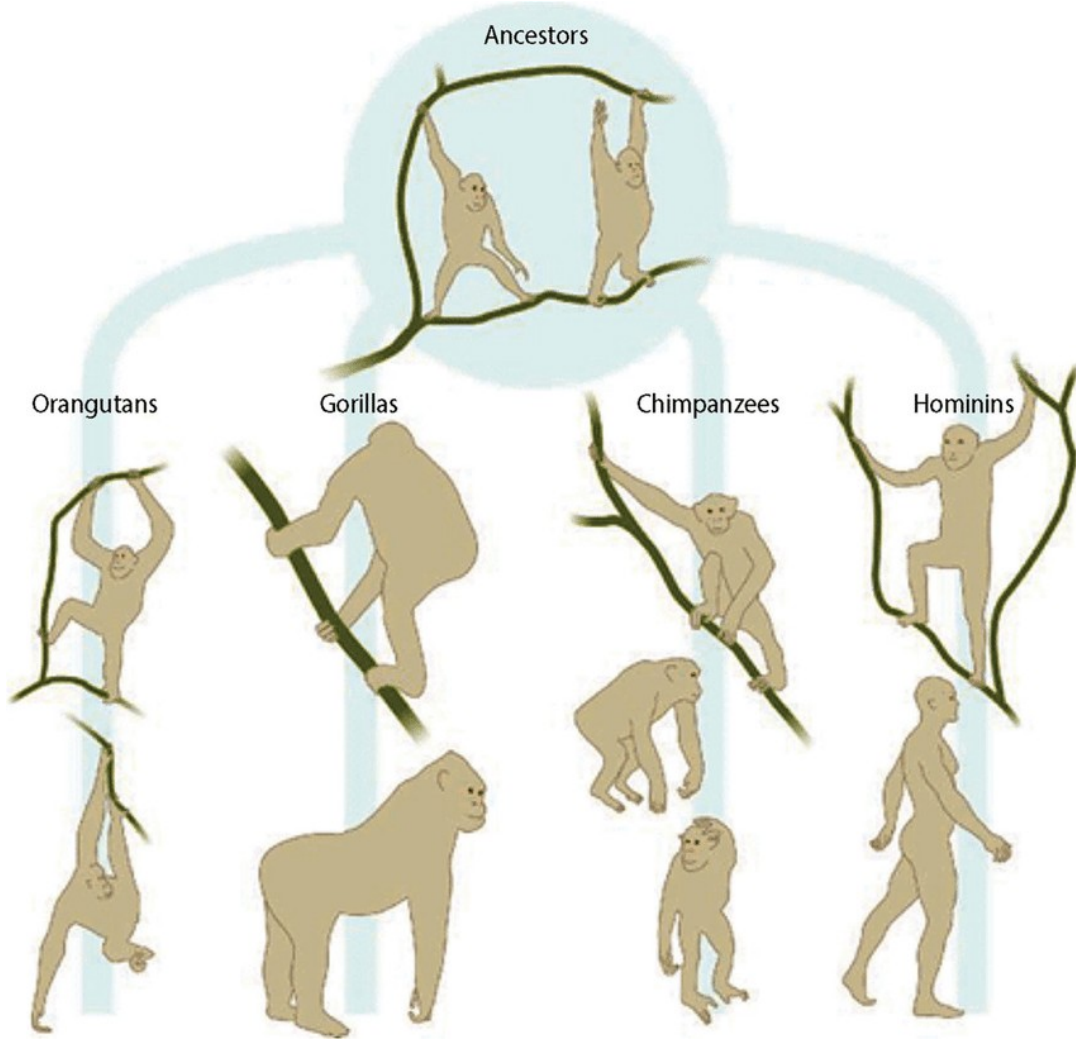
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Post-Dinosaur Extinction Context

- After the dinosaur extinction (66 MYA) mammals diversified
- Primates emerged in newly available arboreal niches
- Behavioral evolution began on a new ecological stage

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Arboreal Constraints

- Life occurred in a three-dimensional canopy
- Falls and footing failures were lethal
- Movement required spatial foresight

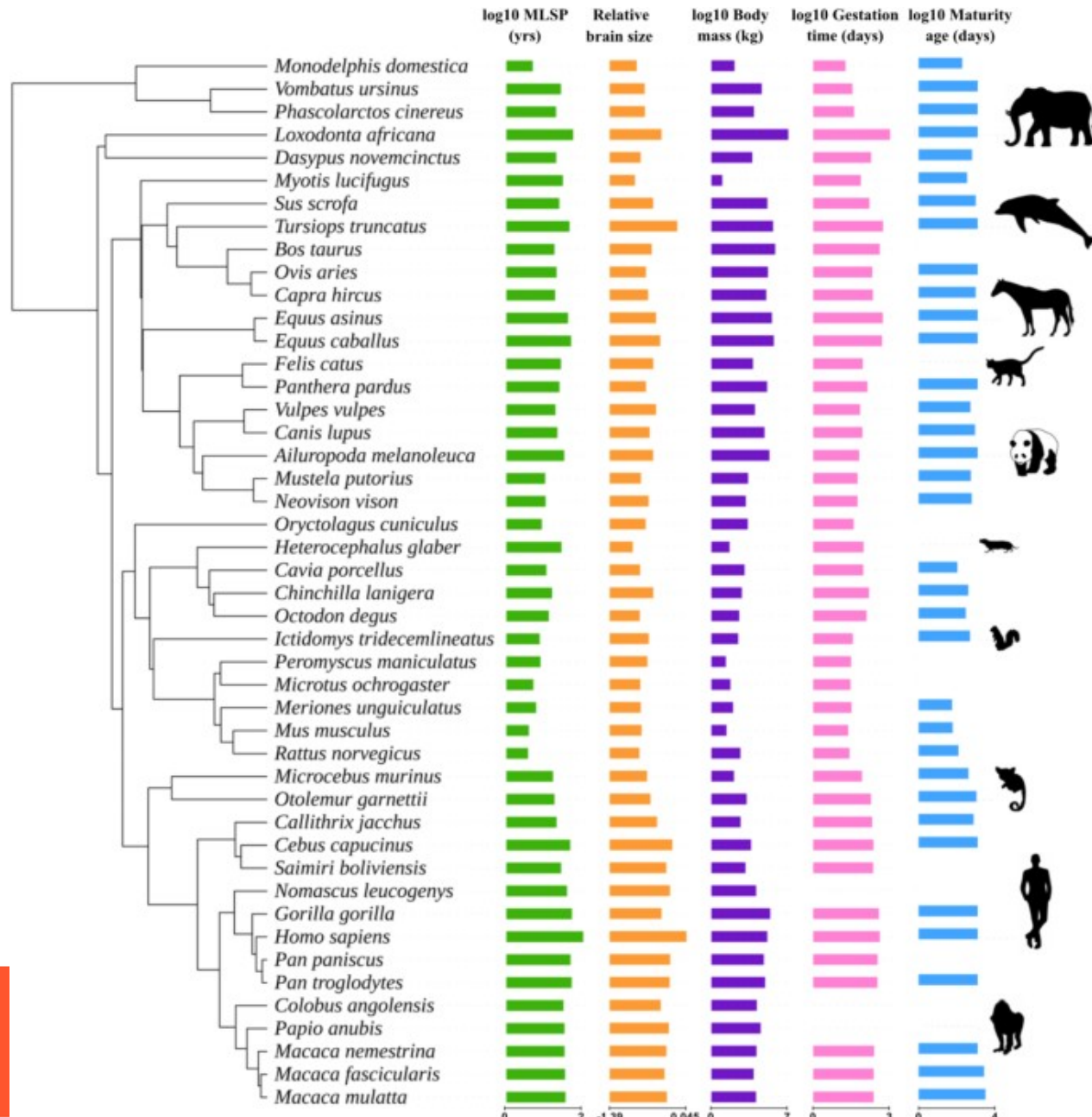
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Predator and Risk Structure

- Predators shaped grouping and vigilance patterns
- Safety depended on others' presence
- Risk was uneven across space and time

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Slow Life Histories and High Cost of Error

- Primates lived long lives for their body size
- They produced few offspring with heavy investment
- Errors had consequences across years

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Stable Groups Under Fission-Fusion Constraints

- Groups persisted over years, not hours
- But subgroups split and rejoin due to travel limits
- Individuals often acted on partial information

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Managing Conflict Without Constant Fighting

- Physical aggression is costly and risky
- Primates use signals and rituals to defuse tension
- Reconciliation repairs relationships after disputes

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Ecological Context	Behavioral Consequences
life in the trees eating fruit variable predators long life large semi-stable groups	?

From Social Structure to Behavioral Demands

- Context created demands for prediction and long-term reasoning
- Nothing here explains the behaviors — it explains why they were needed
- Next we shift to the behavioral phenomena that emerged

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Evolutionary Node	Approx. Date	Best Estimate of Brain Volume
Mammal-Reptile Common Ancestor	~300–320 Mya	~1–3 mL
Mammal Common Ancestor	~180–200 Mya	~2–5 mL
Primate Common Ancestor	~55–65 Mya	~10–15 mL
Ape Common Ancestor	~15–20 Mya	~300–350 mL
Modern Human	present	~1350 mL

1. The Primate Brain Got Much Bigger

- Primates evolved brains far larger than expected for body size
- This enlargement co-occurred with the new behavioral profile
- Bigger brains appeared before we know how they were used



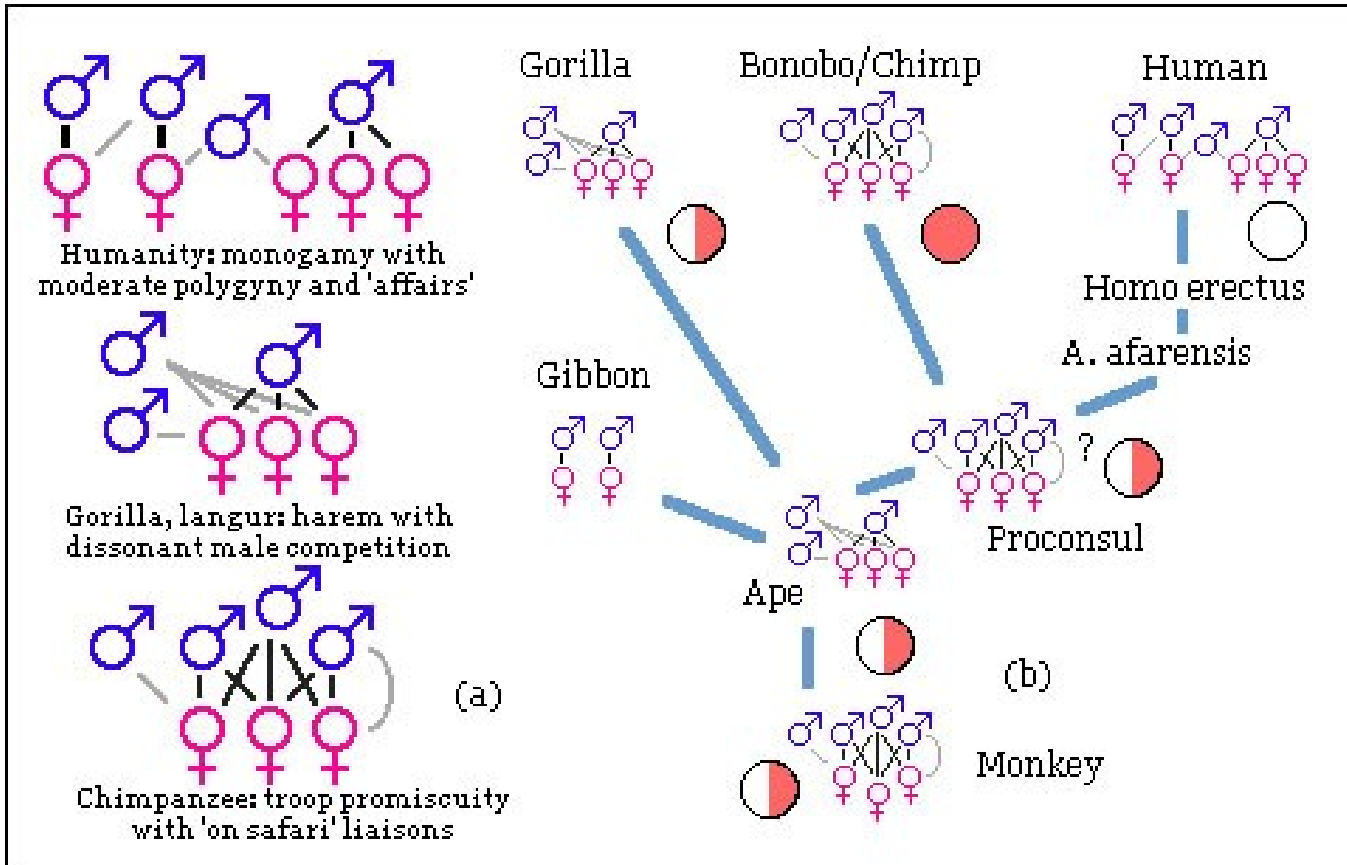
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2. Cooperation Beyond Kin

- Primates cooperated with non-relatives
- They formed and maintained alliances with no shared genes
- This created benefits that could not be reduced to kinship

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3. Stable Social Hierarchies as a Behavioral Regularity

- Primates organized themselves into patterned rank structures
- Rank predicted access to resources, space, allies, and safety
- Hierarchies were long-lived and tracked by all group members

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4. Fairness-Like Responses and Norm Sensitivity

- Primates protest unequal outcomes
- Third parties sometimes react to violations
- Behavior suggests implicit expectations

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5. Long-Horizon Planning and Deferred Payoff Behavior

- Decisions depended on future, not just present
- Actions were taken in expectation of later states
- Payoffs often arrived with delay

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6. Cultural Transmission and Group-Specific Traditions

- Groups differed in stable behavioral practices
- Juveniles acquired local customs over years
- Traditions persisted across generations

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7. Tool Use and Socially Mediated Skill Acquisition

- Some primates use tools routinely
- Tools are learned socially, not reinvented each time
- Complex techniques accumulate within groups

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8. Behavior Appearing to Reflect Internal Emotional Regulation

- Primate behavior changed systematically after social events
- These changes reduced future cost or secured future support
- We observe the pattern, not the internal cause

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9. Ingroup–Outgroup Differentiation

- Primates treated group members and outsiders differently in systematic ways
- The same individual could receive protection or aggression depending on category
- Behavior was organized by membership, not only by personal history

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Partial Parallels in Other Species (and Why They Matter Here)

- Some non-primates show fragments of these behaviors
- No single factor (fruit, sociality, tools) is uniquely primate
- Comparisons prevent oversimplified causal stories

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Why the Parallels Do Not Defeat the Puzzle

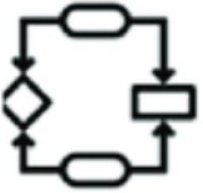
- Having a fragment $\neq$ having the full syndrome
- Correlation does not imply the same causes
- We must guard against simple "X $\Rightarrow$ big brain" claims

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Summary

- Observed a suite of primate behaviors needing explanation
- Identified ecological and social conditions under which they arose

Next Time

- Lecture 2: consider what internal algorithms and representations could produce such behavior.
- Lecture 3: How the primate brain changed, and how it might implement some of these processes